International Journal of Medical Informatics 105 (2017) 22–30

Contents lists available at ScienceDirect

International Journal of Medical Informatics

journal homepage: www.elsevier.com/locate/ijmedinf

Review article

# A systematic review of the effectiveness of interruptive medication prescribing alerts in hospital CPOE systems to change prescriber behavior and improve patient safety

N. Page[a,*], M.T. Baysari[a,b], J.I. Westbrook[a]

[a] Centre for Health Systems and Safety Research, Australian Institute of Health Innovation, Faculty of Medicine and Health Sciences, Macquarie University, Australia
[b] St Vincent's Clinical School, University of NSW, Australia

ARTICLE INFO

*Keywords:*
Medical order entry systems
Computerized provider order entry systems
Electronic prescribing
Decision support systems, clinical
Reminder systems
Medication

ABSTRACT

*Objectives:* To assess the evidence of the effectiveness of different categories of interruptive medication prescribing alerts to change prescriber behavior and/or improve patient outcomes in hospital computerized provider order entry (CPOE) systems.

*Methods:* PubMed, Embase, CINAHL and the Cochrane Library were searched for relevant articles published between January 2000 and February 2016. Studies were included if they compared the outcomes of automatic, interruptive medication prescribing alert/s to a control/comparison group to determine alert effectiveness.

*Results:* Twenty-three studies describing 32 alerts classified into 11 alert categories were identified. The most common alert categories studied were drug-condition interaction (n = 6), drug-drug interaction alerts (n = 6) and corollary order alerts (n = 6). All 23 papers investigated the effect of the intervention alert on at least one outcome measure of prescriber behavior. Just over half of the studies (53%, n = 17) reported a statistically significant beneficial effect from the intervention alert; 34% (n = 11) reported no statistically significant effect, and 6% (n = 2) reported a significant detrimental effect. Two studies also evaluated the effect of alerts on patient outcome measures; neither finding that patient outcomes significantly improved following alert implementation (6%, n = 2).

The greatest volume of evidence relates to three alert categories: drug-condition, drug-drug and corollary order alerts. Of these, drug-condition alerts had the greatest number of studies reporting positive effects (five out of six studies). Only two of six studies of drug-drug interaction and one of six of corollary alerts reported positive benefits.

*Discussion and conclusion:* The current evidence-base does not show a clear indication that particular categories of alerts are more effective than others. While the majority of alert categories were shown to improve outcomes in some studies, there were also many cases where outcomes did not improve. This lack of evidence hinders decisions about the amount and type of decision support that should be integrated into CPOE systems to increase safety while reducing the risk of alert fatigue. Virtually no studies have sought to investigate the impact on changes to prescriber behavior and outcomes overall when alerts from multiple categories are incorporated within the same system.

## 1. Introduction

Most computerized provider order entry (CPOE) systems provide the capacity to enable and configure a number of front-end clinical decision support (CDS) tools to trigger medication-related alerts. Commonly available categories of interruptive medication prescribing alerts include drug-allergy interactions, drug-drug interactions (DDIs), drug-condition interactions, drug-laboratory interactions, dose range checking (DRC), dose adjustment, therapeutic duplication, corollary order, and formulary alerts [1–4]. Local customization may include the ability to activate or deactivate specific alert categories, or allow selection by sub-category, for example, drug-drug interaction (DDI) alerts may be filtered or the alert response tiered according to DDI severity.

Enabling multiple categories of interruptive alerts (e.g. drug-allergy interactions, plus DDIs, plus therapeutic duplication) may have the potential to reduce prescribing error rates further than implementing only one alert category, and it is tempting for hospitals to activate all vendor-supplied CDS alerts on the basis of the more the better.

* Corresponding author.
*E-mail address:* Natalie.Page@hdr.mq.edu.au (N. Page).

http://dx.doi.org/10.1016/j.ijmedinf.2017.05.011
Received 16 December 2016; Received in revised form 17 May 2017; Accepted 19 May 2017
1386-5056/ © 2017 Elsevier B.V. All rights reserved.

However alert fatigue and high rates of alert override are well-recognized consequences of the excessive generation of interruptive alerts; one systematic review reported that 49–96% of medication alerts were overridden by prescribers [5]. Consequently, implementers must be judicious in their decision to include or exclude available interruptive prescribing alerts in their hospital CPOE system. Starting with a small number of well-designed alerts rather than activating all vendor-supplier alerts may maximize clinician buy-in and achieve the desired safety benefits [6–11] and balance the risk of alert fatigue.

While a number of previous systematic reviews have been conducted on CDS in CPOE, and specifically on CPOE alerts, these have generally taken a more aggregated approach, for example looking at multiple care settings or multiple alert types. However, the results from such reviews can less readily be applied to inform decisions about what amount and categories of interruptive medication prescribing alerts should be implemented within a CPOE system.

In this systematic review, we aimed to review the current CPOE literature on the effectiveness of different interruptive medication prescribing alert categories to change prescriber behavior and/or improve patient outcomes. We focused on interruptive medication prescribing alerts in the acute hospital inpatient setting that provide immediate notification of potential errors or safety risks to a prescriber at the time of prescribing.

The purpose of the review is to provide guidance to new implementers on which alert categories have the greatest effectiveness and should be prioritized for implementation in their hospital CPOE system.

## 2. Objectives

Our specific research question was: What is the evidence of the effectiveness of different categories of alert to change prescriber behavior and/or improve patient outcomes in the acute hospital inpatient setting?

## 3. Method

### 3.1. Identification of studies

The PRISMA guideline on the conduct and reporting of systematic reviews was applied [12] and is available as an online data supplement.

A standardized search strategy was developed (Table 1) and applied to the following key bibliographic databases: PubMed, Embase, CINAHL and The Cochrane Library.

### 3.2. Eligibility criteria

Intervention: Studies eligible for inclusion were those undertaken in an acute hospital inpatient setting utilizing a CPOE system with CDS functionality. Studies must have evaluated the impact of automatic, interruptive alert/s that provide immediate notification of potential errors or safety risks to a prescriber at the time of medication order entry. Studies were included if they compared the outcomes from the intervention alert period to either a pre-test period and/or a comparison/control group (e.g. handwritten medication orders or CPOE orders without 'intervention' alert/s) to determine alert effectiveness. The primary outcome measures of interest were: changes in prescriber behavior (e.g. cancelling an order or prescribing a new appropriate order) and/or patient outcomes (e.g. clinical signs or symptoms suggesting potential and/or actual harm).

Exclusions: Articles were excluded if studies were conducted in other practice settings; involved other clinical information systems [13] (e.g. surgery or anesthesia documentation systems, pharmaceutical information system, ICU system); order entry of non-medication interventions (e.g. pathology, radiology); other CDS interventions (e.g. passive decision support; or non-interruptive or asynchronous alerts); if

**Table 1**
Search Strategy.

| Search | Databases | | | |
|---|---|---|---|---|
| | PubMed | Embase | CINAHL[a] | Cochrane |
| 1 | "Medical Order Entry Systems" [MeSH] OR "Hospital Information Systems" [MeSH] OR "Electronic prescribing" [MeSH] OR Electronic presc* | (Electronic AND Prescribing) 'Computerized provider order entry'/exp OR 'Hospital information system'/exp OR 'Electronic prescribing'/exp | MH "Electronic Order Entry" OR MH "Medication Systems" OR MH "Hospital Information Systems" OR MH "Clinical Information Systems +" OR Electronic presc* | "Medical Order Entry Systems" [MeSH] OR"Hospital Information Systems" [MeSH] OR "Electronic prescribing" [MeSH] |
| 2 | "Decision Support Systems, Clinical"[MeSH] OR "Decision Making, Computer-Assisted"[MeSH] OR Deci* | 'Decision support system'/exp OR 'Clinical decision making'/exp OR 'Medical decision making'/exp OR Deci* | MH "Decision Support Systems, Clinical" OR MH "Decision Making, Computer Assisted +" OR Deci* | "Decision Support Systems, Clinical"[MeSH] OR "Decision Making, Computer-Assisted"[MeSH] |
| 3 | "Reminder Systems"[MeSH] OR Remind* OR Alert* OR Prompt* OR Notif* OR Interrupt* | Reminder Systems'/exp OR Alert* OR Prompt* OR Notif* OR Interrupt* | MH "Reminder Systems" OR Alert* OR Prompt* OR Notif* OR Interrupt* | 1 AND 2 |
| 4 | 1 AND 2 AND 3 | 1 AND 2 AND 3 | 1 AND 2 AND 3 | |
| Limits | English Language; Publication 01 January, 2000 to 29 February, 2016 | | | |

[a] Expanders applied included "Apply related words" and "Search within the full text of the articles".

no comparator was included; evaluation outcomes were measured other than those listed in the eligibility criteria; the article did not report results of a primary study; and if the article was only an abstract from conference proceedings with no associated full research paper published.

### 3.3. Data collection process

A standardized data collection form was developed, pilot tested on ten studies and refined. Data abstracted included setting, study design, category of intervention alert and design specifications, other categories of alerts active in the system, and outcomes measured.

### 3.4. Data synthesis

Studies showed considerable variation; there were insufficient studies with similar outcomes and common endpoints to provide comparable quantitative outcome data to conduct a meta-analysis. Instead, data were grouped and synthesised using a narrative approach.

The data abstracted were categorized according to the alert category evaluated. Results were categorised as having (a) a statistically significant beneficial impact on prescriber behavior and/or patient outcomes, (b) a statistically significant detrimental impact, or (c) no statistically significant impact on prescriber behavior and/or patient outcomes (including beneficial or detrimental trend, or no statistical tests performed).

## 4. Results

### 4.1. Study selection

The literature search conducted on 03 March 2016 initially identified a total of 1385 possible articles, 973 after duplicates (n = 412) were removed. A two-stage review process was utilized. In the first stage, one reviewer (NP) screened all titles and abstracts for relevance and eliminated articles if inclusion criteria were not met. A second reviewer (MB) independently appraised 70 random abstracts to ensure consistency in inclusion and exclusion choices.

Following the first stage screening, 60 papers were identified that potentially met the inclusion criteria. Full-text articles were retrieved and reviewed, and two reviewers (NP, MB) carried out the second-stage screening process independently and used a consensus approach to finalize decisions regarding eligibility. A further 40 papers were excluded and 20 eligible studies were identified. In addition to the literature search, 734 potentially relevant articles were identified via hand searching. These articles included the references of systematic reviews and references of the 20 eligible studies. After duplicates were removed (n = 276), titles and abstracts of 458 articles identified in this process were screened and 11 were identified as potentially meeting the inclusion criteria. Following full-text review three studies met the inclusion criteria, bringing the total of studies included in the review to 23 [14–36].

The study selection process can be found in Fig. 1, and the full results of the literature search and the extracted data are available as online data supplements.

### 4.2. Setting

Studies included those undertaken in university hospitals, tertiary hospitals, academic medical centers, and a specialty transplant hospital, with sizes ranging from 90 to 961 beds. Studies were predominantly based in a hospital that had an adult population (n = 19), two in a mixed adult/pediatric setting, and two in a pediatric setting. Patient population was not identified in six studies, and authors were contacted. Three authors responded confirming an adult population; in one paper the adult population was established from the services of the health service named in the article; and in the remaining two an adult population was inferred by the alert specifications (e.g. patient age, dose range settings). Most studies (n = 17) were conducted in a single center.

The prescriber (provider) population was diverse, and sometimes not specified (n = 5). While studies predominantly investigated the impact of medication alerts on the prescribing behaviors of physicians (n = 5 exclusively physicians; n = 13 predominantly physicians), providers also included resident physicians (n = 2), nurse practitioners (n = 5), advance practice nurses (n = 2), and pharmacists (n = 7).

CPOE systems evaluated were predominantly commercial systems (n = 16); the remaining seven studies investigated homegrown systems.

### 4.3. Study design

Study design was classified by the reviewer using the definitions from Harris [37] in order to allow comparisons between studies. Study designs are summarized in Table 2.

### 4.4. Intervention

A variety of alert categories were studied (Table 3). Alert category was classified using the taxonomy described by Wright et al., [3] otherwise according to author definition, or reviewer defined.

In the majority of studies only one category of alert was examined (n = 19), e.g. drug-condition interaction alerts, or DDI alerts (although some studies concurrently investigated other CDS interventions not reported in this review). In the remainder (n = 4), one study investigated an alert suite (specific combination of alerts across multiple categories) comprising of three alert categories (drug allergy interaction, plus DRC, plus duplicate order alerts), and in three studies a suite comprising two alert categories was investigated. In total the 23 papers included in the review reported on 32 alerts across 11 alert categories (Table 3).

We attempted to identify in each study what other alert categories were concurrently active in the CPOE system (but which were not the subject of study), to identify whether the intervention alert/s were part of an alert suite. In the majority of studies (n = 15) it was not stated what (if any) other alert categories were active in the system. In the eight studies that included this information, up to four additional alert categories were described. The most common of these alert suites identified were combinations of two or more of the following alert categories: drug-allergy interaction, DDI, duplicate order, and DRC alerts.

Of the 23 papers, three reported on a new implementation of CPOE, and 20 on the introduction of an alert in an established CPOE system. In total, introduction of one or more new alerts was the focus of 19 papers, and the impact of a modification to an existing alert provided the focus for the remaining four studies.

## 5. Outcomes

### 5.1. Summary of outcomes of studies investigating specific medication alert categories

The outcomes measured and categories of alerts studied are shown in Table 3.

We identified 23 papers that evaluated the impact of alerts on prescriber behavior and/or patient outcomes. These papers described 32 alerts across 11 alert categories. While four studies investigated multiple intervention alerts (i.e. an alert suite) [21–23,29], all 23 papers reported their outcomes specific to the impact of each individual alert category, rather than the impact of an overall alert suite. The rationale for adopting a specific combination of alerts across multiple categories in an alert suite was not described in any papers.

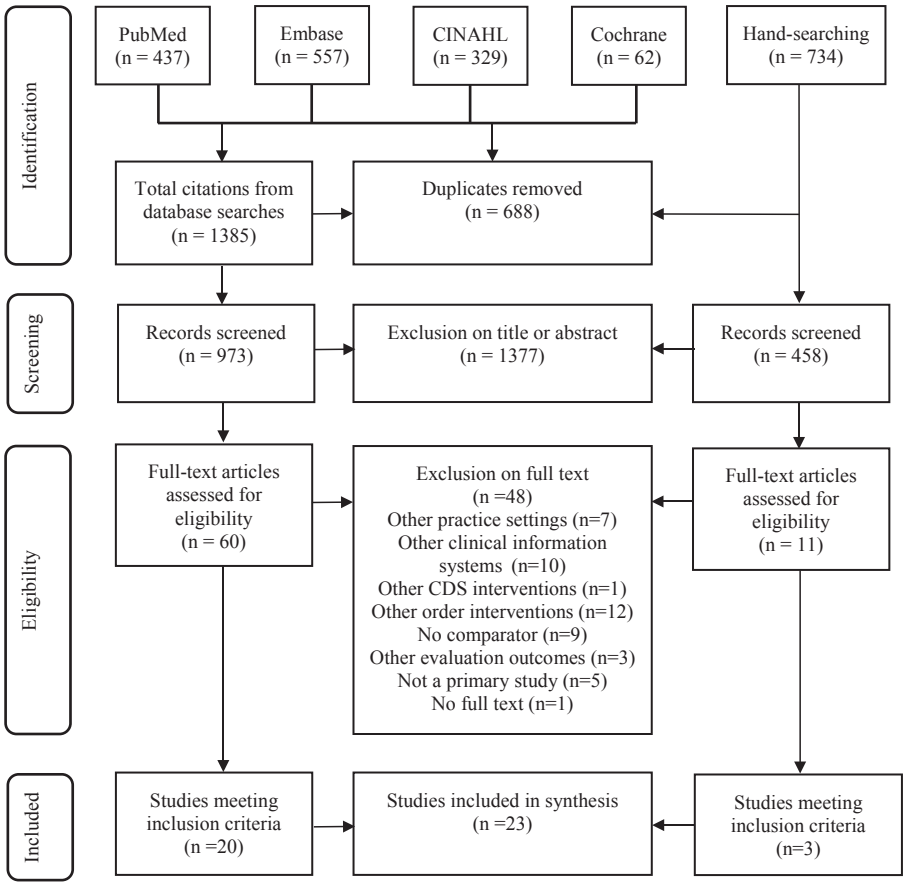



**Fig. 1.** Flow chart showing identification of individual studies for inclusion.

**Table 2**
Design of studies included in review.

| Study Design | Total |
|---|---|
| Prospective | n = 5 |
| Randomized controlled trial (RCT) | 3 |
| Interrupted time-series design | 1 |
| One-Group Pretest-Posttest Design | 1 |
| Retrospective | n = 18 |
| Interrupted time-series design | 2 |
| One-Group Pretest-Posttest Design | 14 |
| Posttest-only design with non-equivalent control group | 1 |
| Repeated-treatment design | 1 |
| Total | n = 23 |

In summary, all 23 papers investigated the effect of an alert on at least one prescriber behavior, reporting on 30 outcome measures of prescriber behavior. Just over half of the studies (53%, n = 17) reported a statistically significant beneficial impact on prescriber behavior as a result of the intervention alert (however one of these also resulted in adverse consequences that necessitated early termination of the study); 34% (n = 11) reported no statistically significant effect, and 6% (n = 2) reported a significant detrimental effect on behavior (increase in errors and override rates).

In addition to investigating the effect on prescriber outcomes, two studies also evaluated the effect of the introduction of two corollary alerts on two patient outcome measures [14,18]; neither studies demonstrated significant improvement (6%, n = 2).

### 5.2. Results by alert category

Results of the effectiveness of alerts were mixed, and there appeared to be little association between a beneficial study outcome and alert category.

As shown in Fig. 2, there were four alert categories where all studies reported a positive impact of alert introduction on prescriber behavior or patient outcomes, however the number of studies investigating these categories was too small to reveal a clinically important effect (formulary alerts n = 3, drug-laboratory interaction alerts n = 1, intravenous fluid alert n = 1, intravenous to oral switch alert n = 1).

The results of two alert categories (drug-condition interaction alerts and DRC alerts) were mixed but studies demonstrated mostly beneficial impact on prescriber behavior or patient outcomes; and the remaining five categories the alerts studied were as likely (or more likely) to result in no significant change or a detrimental effect.

The greatest volume of evidence relates to three alert categories: drug-condition, drug-drug and corollary order alerts. Of these, drug-condition alerts had the greatest number of studies reporting positive effects (five out of six studies). Only two of six studies of drug-drug interaction and one of six of corollary alerts reported positive benefits.

#### 5.2.1. Drug-condition interaction alerts (n = 6)

Of the six studies investigating drug-condition interaction alerts [16,17,24,26,27,31] five studies demonstrated a statistically significant beneficial effect on prescriber behavior, with one study showing no significant difference.

Two studies evaluated new drug-condition interaction alerts specific to patients with, or at risk of, QT interval prolongation. An alert for multiple torsadogenic (QTc interval–prolonging) medications resulted in a 16.8% reduction in completed orders (94% vs. 77%, $p < 0.001$), which translated to a 13.9% decrease in the administration of those medications to patients [31]. Similarly, another paper reported a significant decrease in the rate of inappropriate prescribing of a single torsadogenic medication (intravenous haloperidol) (50% vs. 14%, $p < 0.05$). However, the authors also reported the occasional discontinuation of therapy in patients for who the medication may have

**Table 3**
Intervention alert categories and impact on prescriber behavior and/or patient outcomes.

| Intervention alert categories | Number of times this category of alert was studied[a] | Other alert categories active in CPOE system at the time of study | Reported effect on prescriber behaviors | | | Reported effect on patient outcomes |
|---|---|---|---|---|---|---|
| | | | Number of studies reporting significant beneficial effect from alert | Number of studies reporting no significant change from alert | Number of studies reporting significant detrimental effect from alert | Number of studies reporting no significant impact from alert |
| 1. Drug-condition interaction | 6 | DDI [31]<br>Not stated [16,17,24,26,27] | 5 [17,24,26,27,31] | 1 [16] | | |
| 2. Drug-drug interaction | 6 | Not stated [22,25,28,32,33] | 2 [25,32][b] | 4 [22,28,33][c] | | |
| 3. Corollary order alert | 6 | DDI, DRC, drug-allergy, duplicate [14]<br>Not stated [18,29] | 1 [18][d] | 3 [14,18,29] | | 2 [14,18] |
| 4. Dose range checking | 3 | DRC, drug-allergy, duplicate, dose adjustment [23]<br>DDI,DRC, drug-allergy, [36]<br>Not stated [21] | 2 [23,36] | 1 [21] | | |
| 5. Formulary alert | 3 | DDI, DRC, drug-allergy, duplicate [15]<br>DDI, drug-allergy, duplicate [34]<br>Not stated [19] | 3 [15,19,34] | | | |
| 6. Drug-allergy interaction | 2 | DRC, drug-allergy, duplicate, dose adjustment [23]<br>Not stated [22] | 1 [23] | | 1 [22] | |
| 7. Duplicate order | 2 | DRC, drug-allergy, duplicate, dose adjustment [23]<br>Drug-condition, drug-allergy, duplicate [35] | | 1 [23] | 1 [35] | |
| 8. Dose adjustment | 1 | Not stated [29] | | 1 [29] | | |
| 9. Drug-laboratory alert | 1 | Not stated [30] | 1 [30] | | | |
| 10. Infusion administration checking | 1 | Not stated [21] | 1 [21] | | | |
| 11. Intravenous to oral conversion | 1 | Not stated [20] | 1 [20] | | | |
| Grand Total | **32** | | **17** | **11** | **2** | **2** |

[a] Some papers investigated multiple alert categories, or multiple outcomes, and these were counted as separate studies, hence the number of studies reported in the results is greater than the number of papers included in the review.
[b] Strom et al. [32] significant beneficial effect, however the study was terminated early due to unintended adverse consequences.
[c] Polidori et al. [28] investigated the impact of an individual alert category on two different prescriber outcomes.
[d] Galanter et al. [18] investigated the impact of two alerts from the same category (corollary order alerts) on the same prescriber outcome.

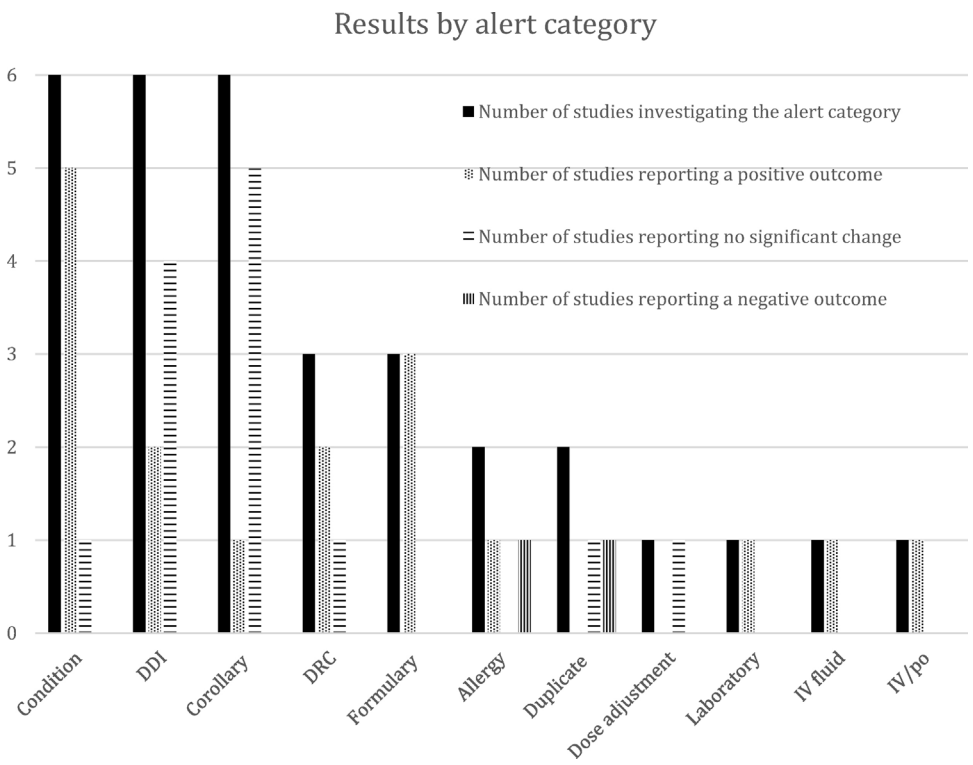



**Fig. 2.** Results by alert category.

been clinically appropriate [26].

New drug-condition interaction alerts targeting hospitalised elderly adults (aged ≥65 years) were examined in three studies. Two institutions designed and implemented alerts aimed at reducing the prescribing of potentially inappropriate medicines using homegrown systems. A significant decrease in the rate of inappropriate prescribing of a group of psychotropic agents (10.8% vs. 7.6%, $p < 0\ 0.001$) [27], and of a subset of the Beers criteria [38] (11.56 vs. 9.94 orders per day, $p < 0.001$) [24] was achieved following the introduction of new alerts. In an RCT evaluating an alert suite to improve the quality of care for older hospitalised patients with cognitive impairment (only one of these alerts meeting the inclusion criteria of this review), introduction of an alert for inappropriate anticholinergic therapy resulted in a non-significant increase in inappropriate orders being cancelled (31.2%–48.9%, $p = 0.11$) [16].

Following introduction of a new drug-condition interaction alert based on a patient's estimated renal function, the likelihood of a patient receiving at least a single dose of a contraindicated medication reduced from 89% to 49% ($p < 0.0001$) [17].

#### 5.2.2. Drug-drug interaction alerts (n = 6)

Six studies investigated drug-drug interaction (DDI) alerts [22,25,28,32,33], all examining prescriber behavior in response to alerts. DDI alerts resulted in a beneficial impact on prescriber behavior in two studies, and a non-significant difference in four studies.

One group conducted two separate RCTs on two customized DDI alerts prompting the prescriber to choose alternative therapy or requiring mandatory acknowledgement of the alert if they chose to continue the order. One study investigated the effectiveness of a redesigned alert using warfarin and NSAID DDIs as a model, and found the alert had no effect on concomitant prescribing [33]. In their other RCT, a new ("nearly hard stop") alert for warfarin and trimethoprim-sulfamethoxazole was designed, requiring an additional acknowledgment of specific clinical diagnosis to proceed with the interacting order. This alert achieved a significant increase in these orders being cancelled compared to the standard practice of a pharmacist intervention program, but the study was terminated early due to unintended adverse consequences: delays in prescribing and administering appropriate medication orders, and failure to prescribe appropriate medication orders [32]. An Italian study also investigated whether requiring mandatory acknowledgement of an existing DDI alert improved alert adherence or the incidence of potential DDIs in an 8-ward acute transplant hospital. Total institutional adherence and DDI error rate were not reported, and no statistical tests were performed, but they did report an increase in alert adherence in 7 wards (reaching over 90% in 6 wards) and a positive trend in the reduction of potential DDIs was observed in four wards [28].

In a study of a homegrown CPOE system, researchers examined compliance with DDI alerts following an alert modification that tiered DDI alerts by severity and made the lowest severity DDI alerts non-interruptive. Compared to the non-tiered control, they found more orders were cancelled in response to tiered alerts (29% vs. 10%, $p < 0.001$) [25]. Following up from a previous study in which they reported high override rates for "critical" DDI alerts, one research team modified their alert to reduce clinically insignificant alerting from topical medication DDIs. This modification resulted in no difference in the number of orders generating a DDI alert (0.25% vs. 0.23%, $p = 0.49$), nor improvement in DDI alert overrides (87% vs. 88%, $p = 0.85$) [22].

#### 5.2.3. Corollary order alerts (n = 6)

Six studies investigated corollary order alerts [14,18,29]. Corollary alerts resulted in a beneficial impact on prescriber behavior in one study, a non-significant impact on prescriber behavior in three studies, and a non-significant impact on patient outcomes in two studies.

Two papers examined the impact of corollary laboratory order alerts on both prescriber behavior and patient outcomes. In a study on corollary laboratory order alerts for aminoglycoside blood-level monitoring in children, introduction of the alert had no impact on the percentage of aminoglycoside courses without aminoglycoside-level monitoring, nor did it have any significant effect on the frequency of therapeutic, subtherapeutic or toxic aminoglycoside levels [14]. In a second paper examining the effect of a corollary order alert in patients receiving digoxin, introduction of an alert increased ordering of

digoxin, magnesium and potassium serum levels but had no significant impact on hypokalemia or hypomagnesemia nor on ordering potassium or magnesium supplementation [18].

A French research team developed a corollary laboratory order alert advising prescribers of the need for an estimated glomerular filtration rate (eGFR) test when prescribing renally cleared and/or nephrotoxic medications. There was no difference in the proportion of orders cancelled in response to the patient not having a current eGFR result before and after the alert was implemented (31.3% vs. 35.0%, p = 0.63) [29].

#### 5.2.4. Dose range checking (n = 3)

In one study a suite of alerts were implemented targeting errors associated with high-risk medicines. The DRC alert resulted in a reduction of excessive dose orders for insulin, but not of other high-risk medications [21]. In another study examining the introduction of CPOE with an alert suite on prescribing errors, a significant reduction in excessive dose errors was reported as a result of a DRC alert (1341 vs. 87, p < 0.001) [23]. A third paper reported a significant, but very small absolute risk reduction, following development of a custom dose checker that alerted prescribers when doses differed from the most common orders for the same medication (0.81% vs. 0.53%, p = 0.015) [36].

#### 5.2.5. Formulary alerts (n = 3)

Alerts targeting prescribing compliance with preferred or formulary medicines were shown to be beneficial in three investigations. In a study of a homegrown system, alerting prescribers to the institution's preferred H2-antagonist increased oral nizatadine prescribing from 15.6% to 81.3% of all H2 orders (p < 0.001) [34]. A second facility targeted eight therapeutic classes of non-formulary agents and enforced therapeutic interchange via a hard-stop alert. Overriding the alert required approval and electronic order entry by a pharmacist. Introduction of the alert resulted in 65% relative reduction in formulary non-adherence (3.5% vs. 1.2%, p < 0.001) [19]. In another study, a formulary alert was effective in reducing the prescribing of a medication subject to supplier shortage (21.5% vs. 9.7%, p < 0.0001), and resulted in a significant increase in the orders for recommended alternatives [15].

#### 5.2.6. Drug-allergy interaction alerts (n = 2)

Two studies on drug-allergy interaction alerts as part of an alert suite were identified. Following implementation of a new CPOE, one study reported a reduction in drug-allergy errors (833 vs. 109 p < 0.001) [23], while another group found drug-allergy alert override increased over time from 69% to 81% (p = 0.005%) [22].

#### 5.2.7. Duplicate order alerts (n = 2)

Reporting on the third alert category within their suite, one research group found no significant impact of a duplicate order alert on duplicate prescribing errors [23]. In a study implementing a new CPOE system in intensive care and critical care units, duplicate errors increased (48 vs. 167 errors; p < 0.0001). Contributing factors were multifactorial, but user interface and visibility of previously ordered and administered medications was considered an issue [35].

#### 5.2.8. Other alert categories (n = 4)

Four papers reported evaluations of alert categories other than those described above. In one study, a dose adjustment alert had no impact on inappropriate prescriptions based on the renal function of inpatients [29]. Introduction of a new intravenous to oral conversion alert for quinolone antibiotics resulted in an overall 5.6% increase in oral orders (p < 0.0001) in a time-series analysis [20]. Implementation of a drug-laboratory interaction alert to reduce inappropriate erythropoietin-stimulating agent use resulted in a relative reduction in inappropriate prescribing of 31.25% (2.82% absolute risk reduction) [30]. And finally, alerts warning when a mandatory dilution fluid was omitted resulted in more dilution fluids being prescribed, with 3584 'dilution fluid omitted' errors prevented over the 6-month post-deployment period [21].

## 6. Discussion

### 6.1. Summary of main results

We found that the most common alert categories studied were drug-condition interactions, drug-drug interaction alerts and corollary order alerts. Just over half of the studies (53%) reported a statistically significant beneficial effect on prescriber behavior from the intervention alert; 34% reported no statistically significant effect, and 6% reported a significant detrimental effect. There was no statistically significant beneficial effect on patient outcome measures (6%, n = 2). This result is less conclusive than a previous systematic review across multiple care settings that reported most alerts (23 out of 27) demonstrated benefit in improving prescribing behavior and/or reducing error rates [39]. This is likely to be because a number of more recent studies have been published which show less favorable results than earlier studies.

While some individual alert categories exclusively or mostly demonstrated benefits (e.g. formulary alerts, drug-condition interaction), there was limited evidence to indicate that any specific category of alerts is more effective than another in changing prescriber behaviors or patient outcomes, to guide new CPOE implementers in prioritizing alerts. The exclusively positive results for the formulary alert category are in contrast to a study examining crude override rates in an ambulatory setting that reported the formulary alert category had the highest average alert override rate of the alert categories examined [40] (followed by age and renal recommendations, categorized as drug-condition alerts in our review; then allergy alerts, DDIs, and duplicate alerts).

No studies identified how combined use of multiple categories of alerts either enhance or detract from overall effectiveness of alerts compared to the implementation of only one alert category within a CPOE system.

A number of factors make it difficult to compare results across studies. First, there was considerable variation in outcome measures, and outcome definitions used (even where similar outcome measures were assessed); as a result, we were unable to extract comparable quantitative outcome data. For example, of the three papers investigating corollary alerts, the three different prescriber outcome measures were: percentage of orders where a laboratory corollary alert was overridden [29]; percentage of orders where no laboratory corollary order was completed [14]; and percentage of prescribers who had ordered an appropriate medication or laboratory corollary order [18]. Recently, leading European researchers published recommendations on relevant outcome measures for evaluating CPOE alerts [11]. Of the 23 papers in our review, only ten measured an outcome recommended by this group.

A second factor that prevented comparisons across studies was alert design. Alert design is proposed to be one of the most significant factors influencing the impact of CDS [41,42], and expert groups have published recommendations on human factors design principles for both alerts in general [43] and relating to specific alert categories [44]. This review reported on alerts from eleven different CPOE systems and although there were variable descriptions of the alert design specifications provided, there appeared to be a lack of common design attributes between systems. Differences between systems included, for example, vendor-specific user-interface features (such as screen display/layout and appearance, terminology); the utilization of a different third party medication knowledge database to support alerting; and local customization. Customization allows alert content to be authored locally, and consequently the alert appearance, its message text, the task context in which it displays, the user to whom it displays, and the response options, *etc.*, can be different even between those sites using the same commercial system, further making comparison difficult. This

configuration is significant as one review found that CPOE customization could have a significant impact on ease of system use, task behavior, and medication errors [45].

### 6.2. Limitations

This systematic review has several limitations. The first is the possibility of exclusion of relevant papers due to the search terms. The terminology used to describe electronic medication order entry systems and interruptive medication prescribing alerts differs between systems, countries and journals. A variety of internationally-appropriate and vendor-agnostic keywords and MeSH headings were utilized within the three theme areas of the search (CPOE; CDS; alerts); and relevant papers were hand-searched; however some relevant papers may have been overlooked. Secondly; the small sample size of eligible studies may limit the conclusions that can be drawn in relation to the effectiveness of different alert categories.

Thirdly, publication bias is likely. That is, negative findings and evaluation and anecdotal experiences and are less likely to have been published and thus the published literature does not necessarily represent the population of completed research or the broad clinician experience. We relied on author reporting to determine whether the intervention alerts were part of an alert suite. Researchers may not have reported on other alerts concurrently active in the CPOE system (but not the subject of the research), and this impacts our ability to draw conclusions about the effectiveness of alert suites.

We limited our review to automatic synchronous interruptive medication alerts displayed to prescribers, as these alerts have the potential to generate excessive alerts and run the risk of alert fatigue and alert override. The advantages and disadvantages of other CDS functionality (such order sentences and order sets), or other alert designs such as non-interruptive alerts, asynchronous alerts or alerts to non-prescribers (e.g. pharmacists) has not been discussed. In some studies, changes observed may have been the result of other CDS functionality in the system not the subject of this review, or other internal or external factors, rather than the medication alerts studied. This was acknowledged as a limitation in three studies [18,19,30]. As alert design can have a significant impact on its success, the generalizability of the results from one system to another is a limitation, as is the ability to infer the results beyond the single center investigated (n = 18); or conclude that results from a homegrown system can be applied to vendor systems.

## 7. Conclusion

This systematic review synthesized the current CPOE literature on the effectiveness of different CPOE interruptive medication prescribing alert categories to change prescriber behavior and/or improve patient outcomes. Just over half of the studies (53%, n = 17) reported a statistically significant beneficial effect from the intervention alert. The majority of alert categories were shown to improve outcomes in some studies, with some individual alert categories exclusively or mostly demonstrating benefits. However, there were also many studies where outcomes did not improve. Virtually no studies sought to investigate the impact on changes to prescriber behavior and outcomes overall when alerts from multiple categories are incorporated within the same system. The current evidence-base does not show a clear indication that particular categories of alerts are more effective than others in hospital CPOE systems. Subsequently organizations are left to make a decision on the amount and type of prescribing alerts to include in hospital CPOE systems with limited evidence to support these decisions.

## Conflict of interest

None.

## Authors contribution

1. Conception or design of the work
Page N, Baysari MT, Westbrook JI
2. Data collection
Page N
3. Data analysis and interpretation
Page N, Baysari MT
4. Drafting the article
Page N, Baysari MT, Westbrook JI
5. Critical revision of the article
Page N, Baysari MT, Westbrook JI
6. Final approval of the version to be published
Page N, Baysari MT, Westbrook JI

Summary table

**What was already known on the topic**

- Interruptive CPOE medication prescribing alerts have the potential to change prescriber behavior and improve patient outcomes
- Alert fatigue and high rates of alert override are well-recognized consequences of the excessive generation of interruptive alerts
- Implementers are faced with the question of what interruptive medication prescribing alerts should be included in CPOE in order to achieve the desired safety benefits and balance the risk of alert fatigue

**What this study added to our knowledge**

- Just over half of the studies reported a statistically significant beneficial impact on prescriber behavior or patient outcomes as a result of the intervention alert
- The majority of alert categories were shown to improve outcomes in some studies, with some individual alert categories exclusively or mostly demonstrating benefits. However, there were also many studies where outcomes did not improve.
- There was limited evidence to indicate that any specific category of alerts is more effective than another in changing prescriber behaviors or patient outcomes.
- Little is known about the impact of implementing a particular combination of alerts across multiple categories on alert burden and decision-support effectiveness overall.

## Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.ijmedinf.2017.05.011.

## References

[1] P.M. Kilbridge, E.M. Welebob, D.C. Classen, Development of the Leapfrog methodology for evaluating hospital implemented inpatient computerized physician order entry systems, Qual. Saf. Health Care 15 (April (2)) (2006) 81–84.
[2] G.J. Kuperman, A. Bobb, T.H. Payne, et al., Medication-related clinical decision support in computerized provider order entry systems: a review, J. Am. Med. Inform. Assoc. 14 (1) (2007) 29–40.
[3] A. Wright, D.F. Sittig, J.S. Ash, et al., Development and evaluation of a comprehensive clinical decision support taxonomy: comparison of front-end tools in commercial and internally developed electronic health record systems, J. Am. Med. Inform. Assoc. 18 (3) (2011) 232–242.
[4] J.A. Osheroff, J.M. Teich, D. Levick, et al., Improving Outcomes with Clinical Decision Support: An Implementer's Guide, second edition, Healthcare Information

and Management Systems Society (HIMSS), Chicago, IL, 2012.
[5] H. Van Der Sijs, J. Aarts, A. Vulto, M. Berg, Overriding of drug safety alerts in computerized physician order entry, J. Am. Med. Inform. Assoc. 13 (2) (2006) 138–147.
[6] D.W. Bates, L.L. Leape, D.J. Cullen, et al., Effect of computerized physician order entry and a team intervention on prevention of serious medication errors, JAMA 280 (15) (1998) 1311–1316.
[7] D.W. Bates, J.M. Teich, J. Lee, et al., The impact of computerized physician order entry on medication error prevention, J. Am. Med. Inform. Assoc. 6 (4) (1999) 313–321.
[8] E. Ammenwerth, P. Schnell-Inderst, C. Machan, U. Siebert, The effect of electronic prescribing on medication errors and adverse drug events: a systematic review, J. Am. Med. Inform. Assoc. (2008) 585–600.
[9] R. Tiwari, D.S. Tsapepas, J.T. Powell, S.T. Martin, Enhancements in healthcare information technology systems: customizing vendor-supplied clinical decision support for a high-risk patient population, J. Am. Med. Inform. Assoc. 20 (2) (2013) 377–380.
[10] D.P. Mulherin, C.R. Zimmerman, B.W. Chaffee, National standards for computerized prescriber order entry and clinical decision support: the case of drug interactions, Am. J. Health Syst. Pharm. 70 (1) (2013) 59–64.
[11] J.J. Coleman, H. van der Sijs, W.E. Haefeli, et al., On the alert: future priorities for alerts in clinical decision support for computerized physician order entry identified from a European workshop, BMC Med. Inform. Decis. Mak. 13 (2013) 111.
[12] A. Liberati, D.G. Altman, J. Tetzlaff, et al., The PRISMA statement for reporting systematic reviews and meta-analyses of studies that evaluate health care interventions: explanation and elaboration, Ann. Intern. Med. 151 (4) (2009) W65–W94.
[13] E. Ammenwerth, N. de Keizer, An inventory of evaluation studies of information technology in health care: trends in evaluation research 1982–2002, Stud. Health Technol. Inform. 107 (Pt. 2) (2004) 1289–1294.
[14] P.A. Abboud, R. Ancheta, M. McKibben, B.R. Jacobs, Impact of workflow-integrated corollary orders on aminoglycoside monitoring in children, Health Inform. J. 12 (September (3)) (2006) 187–198.
[15] B. Bogucki, B.R. Jacobs, J. Hingle, Computerized reminders reduce the use of medications during shortages, J. Am. Med. Inform. Assoc. 11 (July–August (4)) (2004) 278–280.
[16] M.A. Boustani, N.L. Campbell, B.A. Khan, et al., Enhancing care for hospitalized older adults with cognitive impairment: a randomized controlled trial, J. Gen. Intern. Med. 27 (5) (2012) 561–567.
[17] W.L. Galanter, R.J. Didomenico, A. Polikaitis, A trial of automated decision support alerts for contraindicated medications using computerized physician order entry, J. Am. Med. Inform. Assoc. 12 (3) (2005) 269–274.
[18] W.L. Galanter, A. Polikaitis, R.J. DiDomenico, A trial of automated safety alerts for inpatient digoxin use with computerized physician order entry, J. Am. Med. Inform. Assoc. 11 (July–August (4)) (2004) 270–277.
[19] P.J. Helmons, C.R. Coates, J.G.W. Kosterink, C.E. Daniels, Decision support at the point of prescribing to increase formulary adherence, Am. J. Health Syst. Pharm. 72 (5) (2015) 408–413 6p.
[20] T. Hulgan, S.T. Rosenbloom, F. Hargrove, et al., Oral quinolones in hospitalized patients: an evaluation of a computerized decision support intervention, J. Intern. Med. 256 (4) (2004) 349–357.
[21] J. Lee, H. Han, M. Ock, S.I. Lee, S. Lee, M.W. Jo, Impact of a clinical decision support system for high-alert medications on the prevention of prescription errors, Int. J. Med. Inform. 83 (12) (2014) 929–940.
[22] C.P. Lin, T.H. Payne, W.P. Nichol, P.J. Hoey, C.L. Anderson, J.H. Gennari, Evaluating clinical decision support systems: monitoring CPOE order check override rates in the Department of Veterans Affairs' Computerized Patient Record System, J. Am. Med. Inform. Assoc. 15 (September–October (5)) (2008) 620–626.
[23] C.D. Mahoney, C.M. Berard-Collins, R. Coleman, J.F. Amaral, C.M. Cotter, Effects of an integrated clinical information system on medication safety in a multi-hospital setting, Am. J. Health Syst. Pharm. 64 (September (18)) (2007) 1969–1977.
[24] M.L.P. Mattison, K.A. Afonso, L.H. Ngo, K.J. Mukamal, Preventing potentially inappropriate medication use in hospitalized older patients with a computerized provider order entry warning system, Arch. Intern. Med. 170 (15) (2010) 1331–1336.
[25] M.D. Paterno, S.M. Maviglia, P.N. Gorman, et al., Tiering drug-drug interaction alerts by severity increases compliance rates, J. Am. Med. Inform. Assoc. 16 (January–February (1)) (2009) 40–46.
[26] J.M. Pell, D. Cheung, M.A. Jones, E. Cumbler, Don't fuel the fire: decreasing intravenous haloperidol use in high risk patients via a customized electronic alert, J. Am. Med. Inform. Assoc. 21 (Nov (6)) (2014) 1109–1112.
[27] J.F. Peterson, G.J. Kuperman, C. Shek, M. Patel, J. Avorn, D.W. Bates, Guided prescription of psychotropic medications for geriatric inpatients, Arch. Intern. Med. 165 (7) (2005) 802–807.
[28] P. Polidori, C. Di Giorgio, A. Provenzani, Incidence of potential drug interactions in a transplant centre setting and relevance of electronic alerts for clinical practice support, Inform. Prim. Care 20 (4) (2012) 257–262.
[29] E. Sellier, I. Colombet, B. Sabatier, et al., Effect of alerts for drug dosage adjustment in inpatients with renal insufficiency, J. Am. Med. Inform. Assoc. 16 (March–April (2)) (2009) 203–210.
[30] M.S. Selvan, D.F. Sittig, E.J. Thomas, C.C. Arnold, R.E. Murphy, Shabot MM:Improving erythropoietin-stimulating agent administration in a multihospital system through quality improvement initiatives: a pre-post comparison study, J. Patient Saf. 7 (3) (2011) 127–132.
[31] A. Sorita, J.M. Bos, B.W. Morlan, R.F. Tarrell, M.J. Ackerman, P.J. Caraballo, Impact of clinical decision support preventing the use of QT-prolonging medications for patients at risk for torsade de pointes, J. Am. Med. Inform. Assoc. 22 (April (e1)) (2015) e21–e27.
[32] B.L. Strom, R. Schinnar, F. Aberra, et al., Unintended effects of a computerized physician order entry nearly hard-stop alert to prevent a drug interaction: a randomized controlled trial, Arch. Intern. Med. 170 (September (17)) (2010) 1578–1583.
[33] B.L. Strom, R. Schinnar, W. Bilker, S. Hennessy, C.E. Leonard, E. Pifer, Randomized clinical trial of a customized electronic alert requiring an affirmative response compared to a control group receiving a commercial passive CPOE alert: NSAID–warfarin co-prescribing as a test case, J. Am. Med. Inform. Assoc. 17 (July–August (4)) (2010) 411–415.
[34] J.M. Teich, P.R. Merchia, J.L. Schmiz, G.J. Kuperman, C.D. Spurr, D.W. Bates, Effects of computerized physician order entry on prescribing practices, Arch. Intern. Med. 160 (18) (2000) 2741.
[35] T.B. Wetterneck, J.M. Walker, Blosky MA, et al. Factors contributing to an increase in duplicate medication order errors after CPOE implementation, J. Am. Med. Inform. Assoc. 18 (November–December (6)) (2011) 774–782.
[36] A.D. Woods, D.P. Mulherin, A.J. Flynn, J.G. Stevenson, C.R. Zimmerman, B.W. Chaffee, Clinical decision support for atypical orders: detection and warning of atypical medication orders submitted to a computerized provider order entry system, J. Am. Med. Inform. Assoc. 21 (May–June (3)) (2014) 569–573.
[37] A.D. Harris, J.C. McGregor, E.N. Perencevich, et al., The use and interpretation of quasi-experimental studies in medical informatics, J. Am. Med. Inform. Assoc. 13 (1) (2006) 16–23.
[38] D.M. Fick, J.W. Cooper, W.E. Wade, J.L. Waller, J.R. Maclean, M.H. Beers, Updating the Beers criteria for potentially inappropriate medication use in older adults: results of a US consensus panel of experts, Arch. Intern. Med. 163 (22) (2003) 2716–2724.
[39] A. Schedlbauer, V. Prasad, C. Mulvaney, et al., What evidence supports the use of computerized alerts and prompts to improve clinicians' prescribing behavior? J. Am. Med. Inform. Assoc. 16 (July–August (4)) (2009) 531–538.
[40] P.E. Beeler, E.J. Orav, D.L. Seger, P.C. Dykes, D.W. Bates, Provider variation in responses to warnings: do the same providers run stop signs repeatedly? J. Am. Med. Inform. Assoc. 23 (April (e1)) (2016) e93–e98.
[41] J.L. Belden, R. Grayson, J. Barnes, Defining and Testing EMR Usability: Principles and Proposed Methods of EMR Usability Evaluation and Rating, Healthcare Information and Management Systems Society (HIMSS), 2009.
[42] J. Horsky, G.D. Schiff, D. Johnston, L. Mercincavage, D. Bell, B. Middleton, Interface design principles for usable decision support: a targeted review of best practices for clinical prescribing interventions, J. Biomed. Inform. 45 (6) (2012) 1202–1216.
[43] J. Horsky, S. Phansalkar, A. Desai, D. Bell, B. Middleton, Design of decision support interventions for medication prescribing, Int. J. Med. Inform. 82 (June (6)) (2013) 492–503.
[44] T.H. Payne, L.E. Hines, R.C. Chan, et al., Recommendations to improve the usability of drug-drug interaction clinical decision support alerts, J. Am. Med. Inform. Assoc. 22 (November (6)) (2015) 1243–1250.
[45] R. Khajouei, M.W. Jaspers, CPOE system design aspects and their qualitative effect on usability, Stud. Health Technol. Inform. 136 (2008) 309–314.